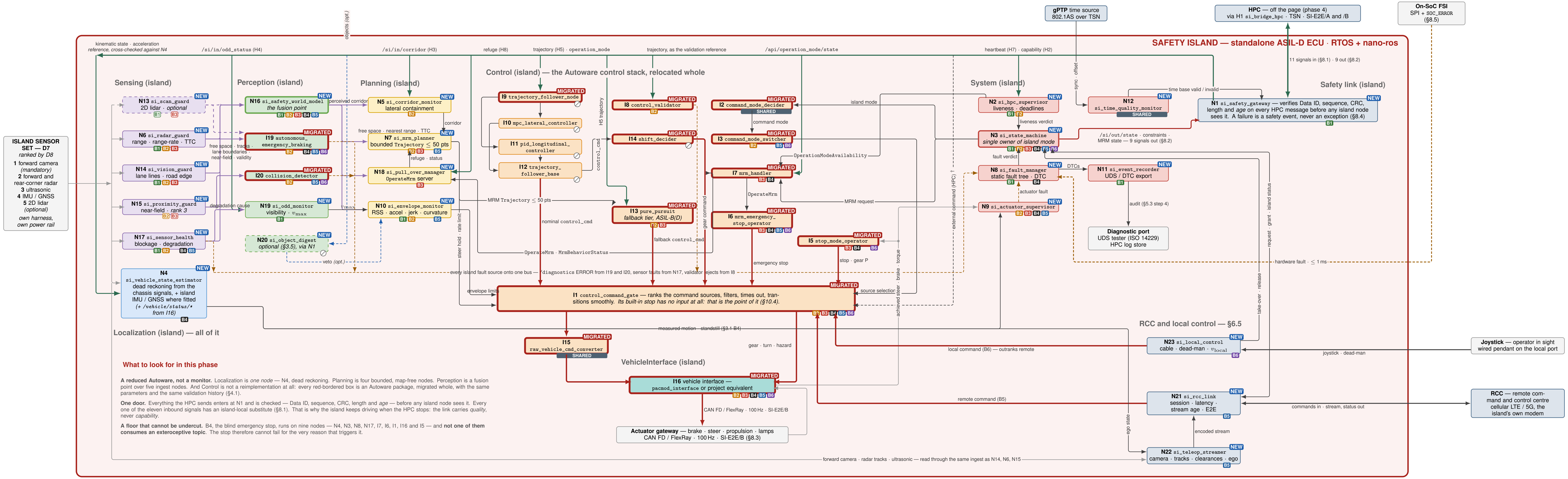
Phase 5 — the Safety Island alone, every node

The last zoom. All 37 island nodes — 18 migrated Autoware **I**-nodes and 19 new **N**-nodes — plus the optional N20, and N21–N23 of §6.5 for B5 and B6 (§6.6).
Colour is still the Autoware stack, as in `ArchitectureMain.pdf`; the island runs a reduced version of nearly every stack. The HPC is off the page: it reaches this diagram only through N1.



SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
gPTP time source 802.1AS over TSN
HPC — off the page (phase 4) via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI SPI + SOC_ERROR (§8.5)
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
sync · offset
11 signals in (§8.1) · 9 out (§8.2)
Sensing (island)
N13 si_scan_guard 2D lidar · optional NEW B1 B3
N6 si_radar_guard range · range-rate · TTC NEW B1 B2 B3
N14 si_vision_guard lane lines · road edge NEW B1 B2 B3
N15 si_proximity_guard near-field · rank 3 NEW B2 B3
N17 si_sensor_health blockage · degradation NEW B1 B2 B4 B5
N4 si_vehicle_state_estimator dead reckoning from the chassis signals, + island IMU / GNSS where fitted (+ /vehicle/status/* from I16) NEW B4
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model the fusion point NEW B1 B2 B3 B4 B5
I19 autonomous_emergency_braking MIGRATED B2 B4 B5 B6
I20 collision_detector MIGRATED B2 B5 B6
N19 si_odd_monitor visibility · v_{max} NEW B1
N20 si_object_digest optional (§3.5), via N1 NEW
free space · tracks · lane boundaries · near-field · validity
degradation cause
perceived corridor
v_{max}
veto (opt.)
Planning (island)
N5 si_corridor_monitor lateral containment NEW B2
corridor
free space · nearest range · TTC
N7 si_mrm_planner bounded Trajectory ≤ 50 pts NEW B2 B3 B5
refuge · status
N18 si_pull_over_manager OperateMrm server NEW B3
N10 si_envelope_monitor RSS · accel · jerk · curvature NEW B1 B2 B5
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node MIGRATED
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
I8 control_validator MIGRATED B2
I14 shift_decider MIGRATED
I13 pure_pursuit fallback tier, ASIL-B(D) MIGRATED B2 B3
MRM Trajectory ≤ 50 pts
nominal control_cmd
fallback control_cmd
gear command
steer hold · rate limit
envelope limits
I2 command_mode_decider MIGRATED SHARED
command mode
I3 command_mode_switcher MIGRATED B2 B5 B6
island mode
OperationModeAvailability
I7 mrm_handler MIGRATED B3 B4
OperateMrm
MRM request
I6 mrm_emergency_stop_operator MIGRATED B3 B4 B5 B6
I5 stop_mode_operator MIGRATED B3 B4 B6
OperateMrm · MrmBehaviorStatus
emergency stop
stop · gear P
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
external command (HPC) †
achieved steer · brake · torque
source selection
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4). MIGRATED B2 B3 B4 B5 B6
measured motion · standstill (§3.1 B4)
I15 raw_vehicle_cmd_converter MIGRATED SHARED
VehicleInterface (island)
gear · turn · hazard
I16 vehicle interface — pacmod_interface or project equivalent MIGRATED B2 B3 B4 B5 B6
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
N2 si_hpc_supervisor liveness · deadlines NEW B1 B2
liveness verdict
N3 si_state_machine single owner of island mode NEW B1 B2 B3 B4 B5 B6
fault verdict
N8 si_fault_manager static fault tree · DTC NEW B1 B4
DTCs
actuator fault
N9 si_actuator_supervisor NEW B2 B3 B4 B5
N12 si_time_quality_monitor NEW SHARED
time base valid / invalid
N11 si_event_recorder UDS / DTC export NEW B1
audit (§5.3 step 4)
Diagnostic port UDS tester (ISO 14229) HPC log store
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4) NEW B1
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
request · grant · island status
hardware fault · ≤ 1 ms
take-over · release
RCC and local control — §6.5
N23 si_local_control cable · dead-man · v_{local} NEW B6
local command (B6) — outranks remote
Joystick — operator in sight wired pendant on the local port
joystick · dead-man
N21 si_rcc_link session · latency · stream age · E2E NEW B5
remote command (B5)
RCC — remote command and control centre cellular LTE / 5G, the island's own modem
commands in · stream, status out
encoded stream
ego state
N22 si_teleop_streamer camera · tracks · clearances · ego NEW B5
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What to look for in this phase

A reduced Autoware, not a monitor. Localization is *one node* — N4, dead reckoning. Planning is four bounded, map-free nodes. Perception is a fusion point over five ingest nodes. And Control is not a reimplementation at all: every red-bordered box is an Autoware package, migrated whole, with the same parameters and the same validation history (§4.1).

One door. Everything the HPC sends enters at N1 and is checked — Data ID, sequence, CRC, length and *age* — before any island node sees it. Every one of the eleven inbound signals has an island-local substitute (§8.1). That is why the island keeps driving when the HPC stops: the link carries *quality*, never *capability*.

A floor that cannot be undercut. B4, the blind emergency stop, runs on nine nodes — N4, N3, N8, N17, I7, I6, I1, I16 and I5 — and **not one of them consumes an exteroceptive topic**. The stop therefore cannot fail for the very reason that triggers it.

Stacks — the colours of `ArchitectureMain.pdf`

- Sensing
- Localization
- Perception
- Planning
- Control
- System
- VehicleInterface
- Vehicle
- Safety link

Where each node came from

- MIGRATED — the same Autoware node, migrated
- NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

- *one fixed slot per behaviour* B1 B2 B3 B4 B5 B6 — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.
- SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.
- B2 B3 — **Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.
- ⊘ — **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.